Regression Calibration

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SER 2026 Short Course

Outline

1. Two versions of regression calibration
2. Regression calibration for the Cox model
3. Pseudo-confounding

Regression Calibration

Analysis goal: to estimate the exposure-outcome association in a regression model (e.g., generalized linear model, Cox model), where the exposure is **continuous** and subject to **non-differential** measurement error.

Model Assumption for the Measurement Error Model

General linear measurement error models are common in epidemiologic studies.

When there are p error-prone variables, the measurement error model could be written as:

$$\mathbf{X_i} = \mathbf{\Gamma_0} + \mathbf{\Gamma_1 Z_i} + \mathbf{\Gamma_2 V_i} + \epsilon_{\mathbf{i}}, \qquad i = 1, ..., n.$$

For subject i, $\mathbf{X_i}$ is a $p \times 1$ vector of true/gold-standard measures, $\mathbf{Z_i}$ is a $p \times 1$ vector of the mismeasured versions of these variables, $\mathbf{\Gamma_1}$ is a $p \times p$ matrix of unknown parameters, $\mathbf{V_i}$ is a $q \times 1$ vector of error-free covariates, and $\mathbf{\Gamma_2}$ is a $p \times q$ matrix of unknown parameters.

Assume this model is aligned with the underlying measurement-error process that produced the observed data.

Assumption for the Validation Study: Transportability

Data from a **validation study**, where both X and Z are available, is required when correcting error under a general linear model.

Transportability: the assumption that the estimates of the measurement error model parameters from the validation study are valid estimates of those representing the measurement error process in the main study. That is, (γ_1, γ_2)-estimates from the validation study are valid estimates of (γ_1, γ_2) in the main study.

Internal validation study that is a random sample of the full study: transportability ✓.

External validation study: transportability **?**

Regression Calibration – Imputation Method (RC-IM)

Intuitively speaking, regression calibration involves using a measurement error model (or called "calibration model") that establishes a relationship between the true (error-free) variable and the observed (error-prone) variable.

This calibration model is then used to make predictions for the true variable in the main study where the true variable is not directly measured.

Book: Measurement Error in Nonlinear Models by R. Carroll, D. Ruppert, L. Stefanski, C. Crainceanu. Chapman and Hall/CRC. 2006

RC-IM under General Linear ME Model

We assume the following linear regression model as the measurement error model:

$$X = \gamma_0 + \gamma_1 Z + \gamma_2 V + \epsilon.$$

Given validation study data, RC-IM follows the following steps:

1. **In the validation study** where all of X, Z and V are available, estimate $\widehat{\gamma}_0$, $\widehat{\gamma}_1$ and $\widehat{\gamma}_2$ from the assumed measurement error model.
2. With the estimated calibration model, and Z, V from the main study, calculate $\widehat{X} = \widehat{\gamma}_0 + \widehat{\gamma}_1 Z + \widehat{\gamma}_2 V$ as the predicted/calibrated value of X for each individual **in the main study**.

3. **In the main study**, regress the outcome Y on $\widehat{X}$ and V with an appropriate model (e.g., linear regression, Poisson regression, logistic regression, and etc.) to obtain estimate of the effect adjusted for measurement error. For example, we may fit the following logistic regression model

$$P(Y=1) = \frac{exp(\beta_0 + \beta_1 \widehat{X} + \beta_2 V)}{1 + exp(\beta_0 + \beta_1 \widehat{X} + \beta_2 V)}$$

and obtain $\widehat{\beta}_1$. This $\widehat{\beta}_1$ is then the estimated log-odds ratio representing the X-Y association adjusting for measurement errors in X.

4. Estimate variances of the corrected effect estimates. **These variances must incorporate additional variation from the estimation of the calibration model itself.** Therefore, variances directly estimated from standard regression model software are biased.

Variance Formula

Variance of $\widehat{\beta}_1$ can be estimated using the sandwich estimator from stacking up for following estimating equations.

Score/estimating equation for the outcome model: $S_1(\beta_0, \beta_1, \beta_2; \gamma) = 0$

Score/estimating equation for the measurement error model: $S_2(\gamma) = 0$

Assumptions for RC-IM under General Linear ME Model

Surrogacy

Measurement errors are non-differential with respect to outcome status.

Surrogacy assumption: $P(Y = 1|X, Z, V) = P(Y = 1|X, V)$ or $Y \perp Z|X, V$

It is guaranteed by the prospective nature of the exposure measurements.

Assumptions for RC-IM under General Linear ME Model

- Surrogacy: Empirically unverifiable unless there are enough cases in the validation study; guaranteed by prospective study design (assuming V includes all W);
- Transportability: Empirically unverifiable
- Linearity in the outcome model: X is included in $\beta_0 + \beta_1 X + \beta_2 V$ as a linear term
- For logistic regression, at least one of the following two:
    - small measurement error variance : $\beta_1^2 \sigma^2_{X|Z,V}$ is small, where $\sigma^2_{X|Z,V}$ is $var(\epsilon)$, the variance of the residual term in the measurement error model.
    - rare D + normally distributed error + homoscedastic error: normally distributed homoscedastic error means ϵ follows Normal distribution with a constant variance.

Why RC-IM works

If we can assume linearity of X in the outcome model:

$$E(Y|X,V) = \beta_0 + \beta_1 X + \beta_2 V.$$

The law of total expectation or the law of iterated expectations:

$$\text{Then,} \qquad \begin{aligned} E(Y|Z,V) &= E_{X|Z,V}\{E(Y|X,Z,V)\} \\ &\overset{\textbf{Surrogacy}}{=} E_{X|Z,V}\{E(Y|X,V)\} \\ &= E_{X|Z,V}\{\beta_0 + \beta_1 X + \beta_2 V\} \\ &= \beta_0 + \beta_1 E(X|Z,V) + \beta_2 E_{X|Z,V}(V) \\ &= \beta_0 + \beta_1 \widehat{X} + \beta_2 V \end{aligned}$$

Example: Dietary Fat Intake and the Risk of Breast Cancer

MS: 89,538 women returned a food frequency questionnaire (FFQ) in 1980 in the Nurses' Health Study (NHS).

- Breast cancer incidence over the first 4 years of the follow-up was ascertained by self-report and validated through medical records review.
- Dietary intake (Z) was measured through a 61-item FFQ asking about the usual frequency of intake of the most common foods during the past year.

VS: 173 women from the MS

- The VS participants recorded 4 one-week diet records (X) based on weighed food intake at 3-month intervals over a one-year period.

Willett WC, Stampfer MJ, Colditz GA, Rosner BA, Hennekens CH, Speizer FE. Dietary fat and the risk of breast cancer. New England Journal of Medicine. 1987 Jan 1;316(1):22-8.

Mean and standard deviation of total fat, total calories, and alcohol intakes in the validation study

	7DDR		FFQ	
	Mean	SD	Mean	SD
Total fat (g/day)	68.6	16.3	56.1	22.0
Total calories (kcal/day)	1620	323.4	1372	482.1
Alcohol (g/day)	9.0	9.7	9.0	12.3

- FFQ underestimates intake of total fat and calories.
- FFQ has more variability.

Correlation between 7DDR and FFQ measures in the validation study

	FFQ		
	Total fat (g/day)	Total calories (kcal/day)	Alcohol (g/day)
7DDR			
Total fat (g/day)	0.37		
Total calories (kcal/day)		0.36	
Alcohol (g/day)			0.85

- Alcohol is well measured.
- Total calories and total fat are poorly measured.

Verification of RC-IM Assumptions Under General Linear ME Model

Surrogacy

Guaranteed by the prospective nature of the exposure assessment.

Possible caveat:

After controlling for confounding, there may remain some residual correlation between the measurement error and unmeasured baseline risk factors for the outcome. Such residual correlation would induce an association of the measurement error with the outcome itself.

Verification of RC-IM Assumptions Under General Linear ME Model (Cont'd)

Transportability

This is an internal validation study from a sampling point of view.

One could quibble about whether the measurement error model estimated from these 173 NHS participants from the greater Boston is representative of the ones that produced the measurement error in the remaining participants across the country.

Verification of RC-IM Assumptions Under General Linear ME Model (Cont'd)

Logistic-linearity of the outcome model

Let's assume this assumption is reasonable. We will use the following logistic regression as the outcome model:

r.h.s. of the model =

fat (10g/day) + calories (800kcal/day) + alcohol (12g/day)

+ I(age∈[40,45)) + I(age∈[45,50)) + I(age∈[50,55)) + I(age>=50)

(age<40 is the reference category for age.)

Verification of RC-IM Assumptions Under General Linear ME Model (Cont'd)

$\{$small measurement error variance$\}$ OR $\{$rare D + normally distributed error + homoscedastic error$\}$

601 cases of breast cancer occurred among 88,937 women during four years of follow-up. The 4-year cumulative incidence of breast cancer was thus 0.7% – the disease is rare! We assume the normal homoscedastic measurement error assumption is satisfied.

Measurement Error Model

$$DR\_Alcohol(12g/day) = FFQ\_Alcohol(12g/day) + FFQ\_Total\ Fat(10g/day)$$
$$+ FFQ\_Calories(800kcal/day) + I(\text{Age in } [40{,}45)) + I(\text{Age in } [45{,}50)) + I(\text{Age in } [50{,}55))$$
$$+ I(\text{Age}{\geq}50) + \epsilon$$
$$DR\_Total\ Fat(10g/day) = FFQ\_Total\ Fat(10g/day) + FFQ\_Alcohol(12g/day)$$
$$+ FFQ\_Calories(800kcal/day) + I(\text{Age in } [40{,}45)) + I(\text{Age in } [45{,}50)) + I(\text{Age in } [50{,}55))$$
$$+ I(\text{Age}{\geq}50) + \epsilon$$
$$DR\_Calories(800kcal/day) = FFQ\_Calories(800kcal/day) + FFQ\_Alcohol(12g/day)$$
$$+ FFQ\_Total\ Fat(10g/day) + I(\text{Age in } [40{,}45)) + I(\text{Age in } [45{,}50)) + I(\text{Age in } [50{,}55))$$
$$+ I(\text{Age}{\geq}50) + \epsilon$$

Regression Calibration – deattenuation Factor method (RC-DF)

The regression calibration based on deattenuation Factor (RC-DF) (Rosner, Spiegelman, Willet, 1990, 1992, American Journal of Epidemiology) is an extension of regression calibration under additional assumptions.

Comparing between RC-IM and RC-DF,

- RC-DF is less computationally intensive.
- for RC-DF, the estimation of the corrected variance, based on the delta method, has a closed formula. The variance of $\widehat{\beta}_1$ from RC-IM is based on the sandwich variance method – this method is implemented in our R function regCalibCRS.

RC-DF under General Linear ME Model

We assume the same linear regression as the measurement error model:

$$X = \gamma_0 + \gamma_1 Z + \gamma_2 V + \epsilon$$

Rosner's method follows the following steps:

1. **In the main study**, regress Y on Z, V to obtain the effect estimates (e.g., $\widehat{\beta}_1^*$). Using the logistic regression as an example:

$$P(Y=1|Z) = \frac{exp(\beta_0^* + \beta_1^* Z + \beta_2^* V)}{1 + exp(\beta_0^* + \beta_1^* Z + \beta_2^* V)}$$

2. **In the validation study**, regress X on Z, V based on the assumed measurement error model and estimate $\widehat{\gamma}_0$, $\widehat{\gamma}_1$ and $\widehat{\gamma}_2$

3. Correct the effect estimate $\widehat{\beta}_1$ for measurement error-caused bias by

$$\widehat{\beta}_1 = \frac{\widehat{\beta}_1^*}{\widehat{\gamma}_1}$$

4. Use the delta method to derive the variance of the effect estimate

$$\widehat{Var}(\widehat{\beta}_1) = \frac{\widehat{Var}(\widehat{\beta}_1^*)}{(\widehat{\gamma}_1)^2} + \frac{(\widehat{\beta}_1^*)^2\widehat{Var}(\widehat{\gamma}_1)}{(\widehat{\gamma}_1)^4}$$

Why RC-DF works

If we can assume linearity for these models:

$$E(X \mid Z, V) = \gamma_0 + \gamma_1 Z + \gamma_2 V, \quad E(Y \mid X, V) = \beta_0 + \beta_1 X + \beta_2 V.$$

$$\begin{aligned}
\text{Then,} \quad E(Y \mid Z, V) &= E_{X|Z,V}\{E(Y \mid X, Z, V)\} \\
&\overset{\text{Sur.}}{=} E_{X|Z,V}\{E(Y \mid X, V)\} \\
&= E_{X|Z,V}\{\beta_0 + \beta_1 X + \beta_2 V\} \\
&= \beta_0 + \beta_1 E(X \mid Z, V) + \beta_2 E_{X|Z,V}(V) \\
&= \beta_0 + \beta_1(\gamma_0 + \gamma_1 Z + \gamma_2 V) + \beta_2 V \\
&= (\beta_0 + \beta_1\gamma_0) + \boldsymbol{\beta_1\gamma_1}\, Z + (\beta_1\gamma_2 + \beta_2)V \\
&= \beta_0^* + \boldsymbol{\beta_1^*}\, Z + \beta_2^* V
\end{aligned}$$

Therefore, $\boldsymbol{\beta_1^* = \beta_1\gamma_1}$ or $\boldsymbol{\beta_1 = \beta_1^*/\gamma_1}$.

Assumptions for RC-DF under General Linear ME Model

- Surrogacy
- Transportability

- Linearity in the outcome model: Z is included in $\beta_0^* + \beta_1^* Z + \beta_2^* V$ as a linear term
- Linearity in the measurement error model **(additional assumption for RC-DF)**: Z is in $X|(Z, V)$ model as a linear term

- For logistic regression: small measurement error **OR** $\{$Rare disease $+$ Normally distributed error $+$ Homoscedastic error$\}$

Verification of RC-DF Assumptions Under General Linear ME Model

The assumptions of **logistic-linearity of the outcome model** and **linearity in the measurement error model**:

One method for investigating non-linearity is to use the restricted cubic spline method for assessing non-linearity.

Violation of Linearity Assumption: Sensitivity Analysis with RC-IM

We found a potentially non-linear relationship for alcohol in the measurement error model. We used RC-IM, which relaxes the linearity assumption, as a sensitivity analysis.

In the VS, we fitted the following non-linear measurement error models:

$$DR\_Alcohol(12g/day) = FFQ\_Alcohol(12g/day) + FFQ\_Total\ Fat(10g/day) + FFQ\_Calories(800kcal/day)$$
$$+ I(\text{Age in [40,45)}) + I(\text{Age in [45,50)}) + I(\text{Age in [50,55)}) + I(\text{Age}\geq 50) + \text{Spline Terms for FFQ\_Alcohol}$$
$$DR\_Total\ Fat(10g/day) = FFQ\_Total\ Fat(10g/day) + FFQ\_Alcohol(12g/day) + FFQ\_Calories(800kcal/day)$$
$$+ I(\text{Age in [40,45)}) + I(\text{Age in [45,50)}) + I(\text{Age in [50,55)}) + I(\text{Age}\geq 50) + \text{Spline Terms for FFQ\_Alcohol}$$
$$DR\_Calories(800kcal/day) = FFQ\_Calories(800kcal/day) + FFQ\_Alcohol(12g/day) + FFQ\_Total\ Fat(10g/day)$$
$$+ I(\text{Age in [40,45)}) + I(\text{Age in [45,50)}) + I(\text{Age in [50,55)}) + I(\text{Age}\geq 50) + \text{Spline Terms for FFQ\_Alcohol}$$

We then estimated the true intakes using the non-linear measurement error models and refitted the standard logistic regression models with these predicted true intakes in the MS.

Compare the Results

Table 1: The Associations Between Total Fat, Total Energy and Alcohol and Risk of Breast Cancer in the NHS

	Uncorrected	Corrected for Measurement Error		
		RC-DF	RC-IM (all linear)	RC-IM (allow non-linear)
Total fat intake (10g/day)	0.98 (0.92, 1.04)	0.93 (0.79, 1.10)	0.93 (0.79, 1.10)	0.96 (0.82, 1.11)
Total energy intake (800 kcal/day)	1.00 (0.77, 1.29)	1.01 (0.45, 2.24)	1.01 (0.44, 2.33)	0.89 (0.41, 1.93)
Alcohol intake (12g/day)	1.15 (1.06, 1.23)	1.24 (1.08, 1.44)	1.24 (1.05, 1.47)	1.27 (1.10, 1.47)

Compare the Results

- After correcting for measurement error in all mismeasured dietary variables, the point estimate of the association between total fat and breast cancer shifts away from the null. However, it remains statistically non-significant.
- The association between alcohol and breast cancer remains highly significant after correcting for measurement error. It suggests that a 1 drink/day increase is associated with a 24% increase in breast cancer risk.
- When linearity is assumed for all nutrients, correcting for the measurement error-caused biases with RC-DF or RC-IM shows the same point estimates with slightly different confidence intervals (Why?)
- Taking into account the non-linear term of alcohol intake in the measurement error model slightly changed the point estimates.

Regression calibration for the Cox model

- In the main study:

$$\{Z_i, V_i, \widetilde{T}_i, \delta_i\},\ i = 1, \cdots, n_m$$

where

  - δ: censoring indicator.
  - $\widetilde{T} = min(T_i, C_i)$, with T_i=event time, C_i=censoring time

- In the internal validation study:

$$\{X_i, Z_i, V_i, \widetilde{T}_i, \delta_i\},\ i = n_m + 1, \cdots, n_m + n_v$$

- In the external validation study:

$$\{X_i, Z_i, V_i\},\ i = n_m + 1, \cdots, n_m + n_v$$

Ordinary regression calibration for the Cox model

(Prentice, R. L. Biometrika 1982)

$$\begin{aligned}&\lambda(t|Z,V)\\ \approx&\lambda_0(t)\exp(\beta_1 E_X[X|Z,V]+\beta_2' V)\end{aligned}$$

The final approximation holds when the event is rare and when one of the following conditions holds:

(1) $X|Z,V$ normal errors with constant variance;

(2) $X|Z,V$ normal errors with small $Var(X|Z,V)\beta_1^2$;

(3) small effect and $X|Z,V$ has constant variance;

(4) small effect and small $Var(X|Z,V)\beta_1^2$.

Risk set regression calibration (RRC)

- Re-calibrating measurement error model within each risk set; not require rare event.
- Xie SX, Wang CY and Prentice RL (2001, JRSSB) developed this for the classical additive measurement error model, $C = c + e$, with replicate data on C in at least a subset, for time-independent exposures, in the Cox model settings.
- Liao XM, Zucker MD, Li Y and Spiegelman D (2011, Biometrics) extended this to time-varying point exposures in the main study/external validation study (MS/EVS) design in the Cox model settings.
- Liao XM, Zhou X, Wang M, Hart JE, Laden F. and Spiegelman D (2018, JRSSC) further extended the RRC method to correct for measurement error in time-varying functions of exposure history in the MS/EVS design in the Cox model settings.

When time-varying exposure is a function of covariate history

Exposure	True	Surrogate
Point	$c_i(t_{ij})$	$C_i(t_{ij})$
Cumulative average	$x_i(t_{ik})$	$X_i(t_{ik})$

- $x_i(t_{ik}) = \frac{\sum_{j=1}^{k} c_i(t_{ij}) I(t_{ij})}{\sum_{j=1}^{k} I(t_{ij})}$ & $X_i(t_{ik}) = \frac{\sum_{j=1}^{k} C_i(t_{ij}) I(t_{ij})}{\sum_{j=1}^{k} I(t_{ij})}$
- $t_{ik} = t_{i1}, \cdots, t_{iN_{1i}}$ age values (our chosen time scale) at which observations of participant i were available
- $I(t_{ij})$ is an indicator function for participant i's exposure being available at age t_{ij}

RRC Formula for Cox Model

In the main study,

$$\begin{aligned}&\lambda(t_{ik};\{C_i(t), t \le t_{ik}\}, \mathbf{Z_i})\\ \approx&\lambda_0(t_{ik})\exp(\beta_2' V_i + \beta_x \mathbb{E}_c[x_i(t_{ik})|\{C_i(t), t \le t_{ik}\}, T_i \ge t_{ik}, V_i])\end{aligned}$$

Note that the final approximation holds when one of the following conditions holds in each distinct risk set $\mathcal{R}_k = \{i : T_i \ge t_{ik}\}$:

(1) $x_i(t_{ik})|\{C_i(t), t \le t_{ik}\}, V_i$ normal errors with constant variance (exact equality holds under this condition);

(2) $x_i(t_{ik})|\{C_i(t), t \le t_{ik}\}, V_i$ normal errors with small $var(x_i(t_{ik})|\{C_i(t), t \le t_{ik}\})\beta_1^2$;

(3) small RR and $x_i(t_{ik})|\{C_i(t), t \le t_{ik}\}, V_i$ has constant variance;

(4) small RR and small $var(x_i(t_{ik})|\{C_i(t), t \le t_{ik}\})\beta_1^2$.

Measurement error model

Suppose we assume that measurement error is the linear regression

$$\mathbb{E}_c\left[c_i(t_{ij}) | C_i(t_{ij}), V_i, t_{ij} \leq T_i\right] = \gamma_0(t_{ij}) + \gamma_1(t_{ij}) C_i(t_{ij}) + \gamma_2(t_{ij}) V_i.$$

Ideally we can estimate $(\gamma_0(t_{ij}), \gamma_1(t_{ij}), \gamma_2(t_{ij}))$ in each validation study risk set $\mathcal{R}_k$ corresponding to each unique failure time in main study and approximately get

$$\text{Cox: } \lambda(t_{ik}; \{C_i(t), t \leq t_{ik}\}, V_i) \approx \lambda_0(t_{ik}) \exp(\beta_2' V_i) exp(\beta_1 \widehat{x}_i(t_{ik}))$$

where

$$\widehat{x}_i(t_{ik}) = \frac{\sum_{j=1}^{k} \{\widehat{\gamma}_0(t_{ij}) + \widehat{\gamma}_1(t_{ij}) C_i(t_{ij}) + \widehat{\gamma}_2(t_{ij}) V_i)\} I(t_{ij})}{\sum_{j=1}^{k} I(t_{ij})}$$

Principles of Modeling

In multivariate regression models, perfectly measured covariate coefficients will be biased if they are correlated with mis-measured model covariates.

In particular, if there is only one variable measured with error, X, and one measured without error, V,

$$\beta_2^* = \beta_2 + \beta_1 \frac{\rho_{XV}(\sigma^2/\sigma_X^2)(\sigma_X/\sigma_V)}{1-\rho_{XV}^2}$$

where $\sigma_X^2 = Var(X)$, $\sigma_V^2 = Var(V)$, and $\rho_{XV} = Corr(X, V)$.

$\beta_2 =$ the coefficient of V in the model $Y|X, V$ (corrected outcome model).

$\beta_2^* =$ the coefficient of V in the model $Y|Z, V$ (uncorrected outcome model).

(The formula above is from Rosner et al., 1992.)

Pseudo-confounding

Even if V is not a risk factor of the outcome, and hence not a confounder, if it is correlated with X (and presumably with Z), it will appear to be a risk factor in the uncorrected primary regression model, i.e.

Since as given in the previous slide

$$\beta_2^* = \beta_2 + \beta_1 \frac{\rho_{XV}(\sigma^2/\sigma_X^2)(\sigma_X/\sigma_V)}{1-\rho_{XV}^2}$$

Even when $\beta_2 = 0$, $\beta_2^* = \beta_1 \frac{\rho_{XV}(\sigma^2/\sigma_X^2)(\sigma_X/\sigma_V)}{1-\rho_{XV}^2} \neq 0$ unless $\rho_{XV} = 0$

Tests under Multiple Linear Regression With Error

- Model

$$Y = \beta_0 + \beta_1 X + \beta_v V + e$$
$$Z = X + \widetilde{\epsilon}$$

- Global null test is OK: Naive test of $H_0 : \beta_1 = 0$ and $\beta_v = 0$ is valid

- Naive test of $H_0 : \beta_1 = 0$ is valid

What Naive Tests are Invalid?

- Tests for V: surprisingly, tests about the variables V measured without error are not always valid
  - Exception is when X and V are independent

- When X is multivariate: Naive tests here for single components of X are not always valid

Possible Solutions

- Ignore data on V in both the main study and validation study.
- Include V in both the main study and validation study regressions. Since we typically don't know if V is truly a risk factor or not, this is typically the 'safer' option.

Advanced Topic 1: Which V to include in the outcome model and the measurement error model

Wenze Tang, Donna Spiegelman, Xiaomei Liao, Molin Wang*. Causal selection of covariates in regression calibration for mismeasured continuous exposure. Epidemiology. 2024 May 1;35(3):320-8.

Wenze Tang, Donna Spiegelman, Yujie Wu, Molin Wang*. Causal Covariate Selection for the Regression Calibration Method for Exposure Measurement Error Bias Correction. Statistics in Medicine. 2026 Feb;45(3-5):e70430.

Covariates Adjustment Strategies

Decisions need to be made on whether a covariate should be adjusted only in the disease model, or only in the measurement error model, or in both the disease and measurement error models.

Questions:

(1) For validity, what is the minimal covariate set to adjust for in the regression calibration method.

(2) For better efficiency, how to adjust for a given covariate.

Four Possible Covariate Adjustment Strategies under Linear Mean Models

- Deattenuation Factor Method (DF)
  1. $E(X) = \gamma_0 + \gamma_1 Z$ vs $E(X) = \gamma_0 + \gamma_1 Z + \gamma_2 V$
  2. $E(Y) = \beta_0^* + \beta_1^* Z$ vs $E(Y) = \beta_0^* + \beta_1^* Z + \beta_2^* V$
- Imputation Method (IM)
  1. $E(X) = \gamma_0 + \gamma_1 Z$ vs $E(X) = \gamma_0 + \gamma_1 Z + \gamma_2 V$
     $\rightarrow \widehat{X} = \widehat{\gamma}_0 + \widehat{\gamma}_1 Z$ vs $\widehat{X} = \widehat{\gamma}_0 + \widehat{\gamma}_1 Z + \widehat{\gamma}_2 V$
  2. $E(Y) = \beta_0 + \beta_1 \widehat{X}$ vs $E(Y) = \beta_0 + \beta_1 \widehat{X} + \beta_2 V$

For each covariate V, we can:

ME Model	Outcome Model
✓	✓
✗	✗
✓	✗
✗	✓

Key Question: what covariates should we collect and how to adjust for these covariates in regression calibration models?

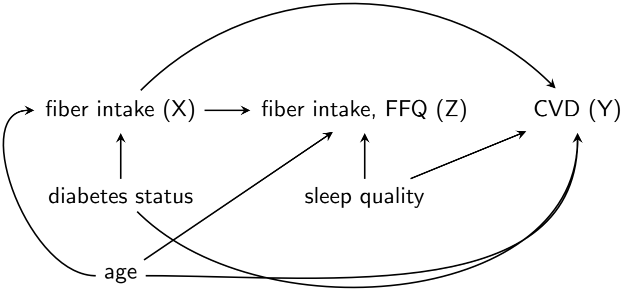
- DF in practice: the same covariate set in MEM and outcome model.

Key Question: what covariates should we collect and how to adjust for these covariates in regression calibration models?

- DF in practice: the same covariate set in MEM and outcome model.
- IM in practice: data-driven covariate selection for MEM:
  1. Huang, 2014: R^2 as selection criteria
  2. Prentice, 2017: $p < 0.1$ as selection criteria
  3. Prentice, 2019: true intake $\sim$ biomarkers $+$ participant characteristics $+$ potential confounders; R^2 as selection criteria

Motivating Example 1: fiber intake and risk of cardiovascular disease (CVD)

Figure 1: "Typical" Measurement Error Process (Hernan and Cole 2009)



Questions that can be asked:

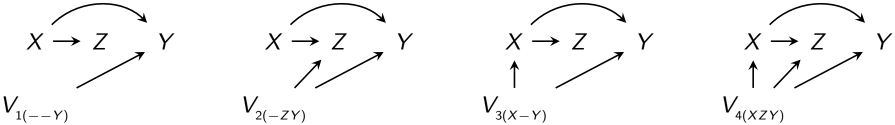
→ 1. Should sleep quality be adjusted in both MEM and outcome model, for both DF and IM methods?

→ 2. Should confounders such as diabetes status be adjusted in the MEM of IM method?

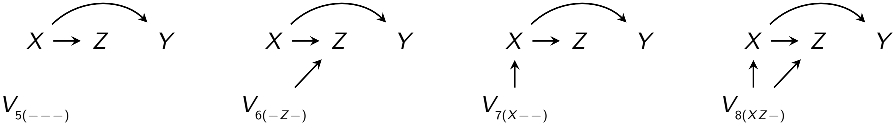
→ 3. Will adjusting for additional covariates in either MEM or outcome model increase efficiency?

Directed Acyclic Graphs(DAGs) and Assumptions

Figure 2: DAGs Illustrating Common Covariate Structures



(a) DAG 1 (b) DAG 2 (c) DAG 3 (d) DAG 4



(e) DAG 5 (f) DAG 6 (g) DAG 7 (h) DAG 8

Results of RC-DF Estimator

	OM	$--$	$-M$	$O-$
DAG 1, $(--Y)$	✓(efficient)	✓	✓	✓(efficient)
DAG 2, $(-ZY)$	✓	biased	biased	biased
DAG 3, $(X-Y)$	✓	biased	biased	biased
DAG 4, (XZY)	✓	biased	biased	biased
DAG 5, $(---)$	✓	✓	✓	✓
DAG 6, $(-Z-)$	✓(efficient)	✓	biased	biased
DAG 7, $(X--)$	✓	✓(efficient)	biased	biased
DAG 8, $(XZ-)$	✓[b]	✓[b]	biased	biased

[b] Relative efficiency depends on strength and direction of $\rho_{V,X}$, $\rho_{X,Z|V}$, $\rho_{V,Z|X}$ and $\rho_{X,Y|V}$, as well as the sample size of main study and validation study.

Results of RC-IM Estimators

	OM	$--$	$-M$	$O-$
DAG 1, $(--Y)$	✓(efficient)	✓	✓	✓(efficient)
DAG 2, $(-ZY)$	✓(efficient)	biased	✓	biased
DAG 3, $(X-Y)$	✓	biased	biased	biased
DAG 4, (XZY)	✓	biased	biased	biased
DAG 5, $(---)$	✓	✓	✓	✓
DAG 6, $(-Z-)$	✓(efficient)	✓	✓(efficient)	biased
DAG 7, $(X--)$	✓	✓	✓(efficient)	biased
DAG 8, $(XZ-)$	✓	✓	✓(efficient)	biased

Summary of Main Results

- Caution about data-driven covariate selection approach for building MEM.
- For both DF and IM methods, risk factors that are correlated with X and/or measurement error need to be adjusted for in both ME and outcome model.
  - Per simulation results, when they are not available in the validation study, they should still be adjusted for at least in the outcome model only, provided that these covariates are not strongly correlated with the measurement error.

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- Caution about data-driven covariate selection approach for building MEM.
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- Efficiency gained when adjusting for risk factors that are uncorrelated with X and Z in the outcome model for continuous outcome under linear model but not necessary for binary outcome under logistic model.

Summary of Main Results

- Non-risk factors that are associated with the true exposure:
  - For DF method, leave it out from either model for better efficiency.
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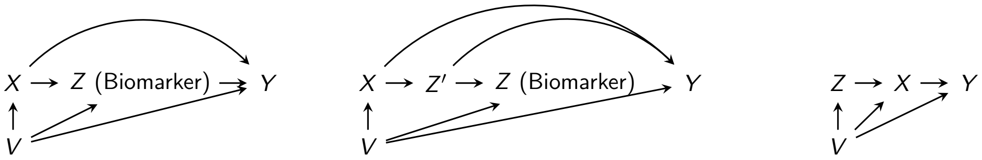
    - For DF method, include it in both models for better efficiency.
    - For IM method, adjusted in MEM only for better efficiency.
- Non-risk factors that are associated with both the true exposure and the ME:
    - For DF method, include it in both models or neither model.
    - For IM method, adjusted in MEM only for better efficiency; IM>DF.

Other Measurement Error Processes

Figure 3: Other Measurement Error Structures



(a) (b) (c)